1 VECTOR FIELDS AND DEL OPERATOR

Scalar field: $\phi(x, y, z)$

$$\mathbf{F}(x, y, z) = P(x, y, z)\,\mathbf{i} + Q(x, y, z)\,\mathbf{j} + R(x, y, z)\,\mathbf{k}$$

$$\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}$$

∇ is a vector differential operator. Its three actions produce the three basic operators: $\nabla\phi$ **(gradient)**, $\nabla\cdot\mathbf{F}$ **(divergence, dot product)**, $\nabla\times\mathbf{F}$ **(curl, cross product)**. Components of $\mathbf{F}$ and ϕ are assumed continuously differentiable.

2 GRADIENT scalar → vector

$$\nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

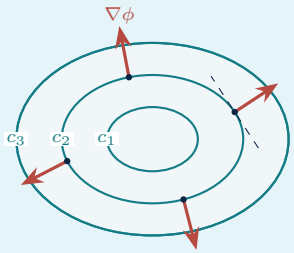
Directional derivative ($\mathbf{u}$ a *unit* vector):

$$D_{\mathbf{u}}\phi = \nabla\phi\cdot\mathbf{u}$$

Maximum rate of increase and its direction ($\nabla\phi \neq \mathbf{0}$):

$$\max D_{\mathbf{u}}\phi = |\nabla\phi|, \qquad \mathbf{u} = \frac{\nabla\phi}{|\nabla\phi|}$$

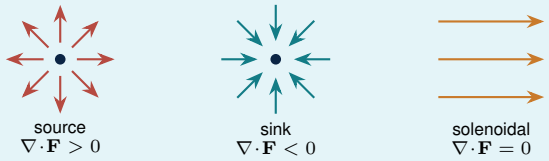
- $\nabla\phi$ points in the direction of **steepest ascent**
- $-\nabla\phi$: steepest descent
- $\nabla\phi \perp$ level curve / surface $\phi = c$, so $\nabla\phi/|\nabla\phi|$ is a unit normal



level curves $\phi = c$

3 DIVERGENCE vector → scalar

$$\nabla\cdot\mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$



- Measures the local **net outward flux density**
- $\nabla\cdot\mathbf{F} > 0$: source-like · < 0: sink-like
- $\nabla\cdot\mathbf{F} = 0$: **solenoidal** field
- Example: $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \;\Rightarrow\; \nabla\cdot\mathbf{F} = 3$

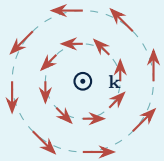
4 CURL vector → vector

$$\nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

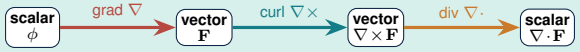
Expanded form:

$$\nabla\times\mathbf{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\mathbf{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\mathbf{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}$$

- Measures local **rotational tendency**
- **Irrotational** field: $\nabla\times\mathbf{F} = \mathbf{0}$
- Plane field $P\mathbf{i} + Q\mathbf{j}$: curl $= \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathbf{k}$
- $\mathbf{F} = -y\mathbf{i} + x\mathbf{j} \;\Rightarrow\; \nabla\times\mathbf{F} = 2\mathbf{k}$



Operator chain: each consecutive pair composes to zero



$\nabla\times(\nabla\phi) = \mathbf{0}$ and $\nabla\cdot(\nabla\times\mathbf{F}) = 0$

5 IMPORTANT VECTOR IDENTITIES

First-order identities

$$\begin{aligned}
\nabla(\phi+\psi) &= \nabla\phi + \nabla\psi \\
\nabla(\phi\psi) &= \phi\nabla\psi + \psi\nabla\phi \\
\nabla\cdot(\phi\mathbf{F}) &= \nabla\phi\cdot\mathbf{F} + \phi\nabla\cdot\mathbf{F} \\
\nabla\times(\phi\mathbf{F}) &= \nabla\phi\times\mathbf{F} + \phi\nabla\times\mathbf{F}
\end{aligned}$$

Second-order identities (components twice continuously differentiable)

$$\begin{aligned}
\nabla\cdot(\nabla\phi) &= \nabla^2\phi \\
\nabla\times(\nabla\phi) &= \mathbf{0} \\
\nabla\cdot(\nabla\times\mathbf{F}) &= 0 \\
\nabla\times(\nabla\times\mathbf{F}) &= \nabla(\nabla\cdot\mathbf{F}) - \nabla^2\mathbf{F}
\end{aligned}$$

Laplacian:

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

$\nabla^2\mathbf{F}$ is the Laplacian applied to each component of $\mathbf{F}$.

Notation used in this sheet

$\mathbf{i}, \mathbf{j}, \mathbf{k}$ unit vectors · $\mathbf{n}$ unit normal · $\mathrm{d}s$ arc length · $\mathrm{d}S$ surface element · $\mathrm{d}A$ area element · $\mathrm{d}V$ volume element · $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ position vector.

Quick check

$\phi = x^2y + z^2$ at $(1, 1, 1)$: $\nabla\phi = (2xy,\ x^2,\ 2z) = (2, 1, 2)$, $|\nabla\phi| = 3$, so the maximum rate of increase is 3 along $\frac{1}{3}(2\mathbf{i} + \mathbf{j} + 2\mathbf{k})$.

$\mathbf{F} = xy\,\mathbf{i} + yz\,\mathbf{j} + zx\,\mathbf{k}$: $\nabla\cdot\mathbf{F} = x + y + z$, $\nabla\times\mathbf{F} = -y\mathbf{i} - z\mathbf{j} - x\mathbf{k}$.

6 LINE INTEGRALS

Scalar: $\displaystyle\int_C \phi\,\mathrm{d}s$ **Vector:** $\displaystyle\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r}$

where $\mathrm{d}\mathbf{r} = \mathrm{d}x\,\mathbf{i} + \mathrm{d}y\,\mathbf{j} + \mathrm{d}z\,\mathbf{k}$. For $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k},\ a \le t \le b$:

$$\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,\mathrm{d}t$$

$$\int_C \phi\,\mathrm{d}s = \int_a^b \phi(\mathbf{r}(t))\,|\mathbf{r}'(t)|\,\mathrm{d}t$$

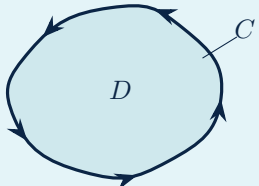
The vector line integral is the **work done by a force field $\mathbf{F}$** along C. Reversing the direction of C changes its sign.

7 GREEN'S THEOREM

For a **positively oriented**, simple, closed curve C enclosing a plane region D:

$$\oint_C (P\,\mathrm{d}x + Q\,\mathrm{d}y) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\mathrm{d}A$$

- C piecewise smooth, simple, closed
- P, Q have continuous first partials on an open set containing $D \cup C$
- Positive orientation: D lies on the left
- Line integral ⇒ double integral
- Area: $A = \frac{1}{2}\oint_C (x\,\mathrm{d}y - y\,\mathrm{d}x)$



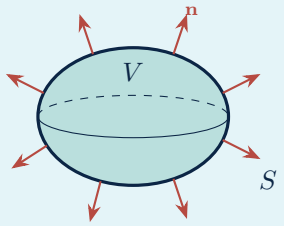
positive (counter-clockwise) orientation: D on the left

Check: C: $x^2 + y^2 = 1$, $P = -y$, $Q = x$ ⇒ $\oint_C (-y\,\mathrm{d}x + x\,\mathrm{d}y) = \iint_D 2\,\mathrm{d}A = 2\pi$.

8 GAUSS DIVERGENCE THEOREM

$$\oiint_S \mathbf{F}\cdot\mathbf{n}\,\mathrm{d}S = \iiint_V \nabla\cdot\mathbf{F}\,\mathrm{d}V$$

- S: **closed**, piecewise smooth surface bounding the solid V
- $\mathbf{n}$: **outward** unit normal
- $\mathbf{F}$ has continuous first partials on $V \cup S$
- $\oiint_S \mathbf{F}\cdot\mathbf{n}\,\mathrm{d}S$ is the net outward flux
- Volume: $V = \frac{1}{3}\oiint_S \mathbf{r}\cdot\mathbf{n}\,\mathrm{d}S$



$\mathbf{n}$: outward from V

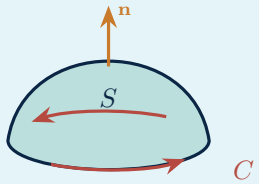
Check: $\mathbf{F} = \mathbf{r}$, V unit ball ⇒ flux $= \iiint_V 3\,\mathrm{d}V = 4\pi$.

Surface flux = volume integral of divergence. If V has cavities, include every boundary surface with $\mathbf{n}$ pointing out of V.

9 STOKES' THEOREM

$$\oint_C \mathbf{F}\cdot\mathrm{d}\mathbf{r} = \iint_S (\nabla\times\mathbf{F})\cdot\mathbf{n}\,\mathrm{d}S$$

- S: open, piecewise smooth, oriented surface; C: its boundary curve
- Orientation by the **right-hand rule**
- $\mathbf{F}$ has continuous first partials on an open set containing S
- Circulation around C = flux of the curl through S
- Green's theorem is the case $S \subset xy$-plane, $\mathbf{n} = \mathbf{k}$



right-hand rule: thumb along $\mathbf{n}$, fingers along C

Check: $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$, S unit disc, $\mathbf{n} = \mathbf{k}$ ⇒ $\iint_S 2\,\mathrm{d}S = 2\pi = \oint_C \mathbf{F}\cdot\mathrm{d}\mathbf{r}$.

10 CONSERVATIVE VECTOR FIELDS

$\mathbf{F} = \nabla\phi$, where ϕ is a **scalar potential**.

$$\int_C \mathbf{F}\cdot\mathrm{d}\mathbf{r} = \phi(B) - \phi(A), \qquad \oint_C \mathbf{F}\cdot\mathrm{d}\mathbf{r} = 0$$

Test: if $\mathbf{F}$ is C^1 on a **simply connected** domain,

$$\nabla\times\mathbf{F} = \mathbf{0} \iff \mathbf{F} = \nabla\phi$$

Domain matters: $\mathbf{F} = \dfrac{-y\mathbf{i} + x\mathbf{j}}{x^2 + y^2}$ has $\nabla\times\mathbf{F} = \mathbf{0}$ on $\mathbb{R}^2 \setminus \{(0,0)\}$, yet $\oint_C \mathbf{F}\cdot\mathrm{d}\mathbf{r} = 2\pi$ on the unit circle.

11 QUICK COMPARISON TABLE

Operator/ Theorem	Mathematical Operation	Interpretation
Gradient	Scalar → Vector	Rate and direction of increase
Divergence	Vector → Scalar	Local net outward flux density
Curl	Vector → Vector	Local rotational tendency
Green's theorem	Line ↔ Double integral	Circulation in a plane
Gauss theorem	Surface ↔ Volume integral	Flux
Stokes' theorem	Line ↔ Surface integral	Circulation

12 COMMON ERRORS — QUICK CHECK

- Confusing divergence with curl
- Forgetting the unit vector in a directional derivative
- Using a non-unit vector in $D_{\mathbf{u}}\phi = \nabla\phi\cdot\mathbf{u}$
- Wrong determinant expansion in the curl (the $\mathbf{j}$ term has a sign flip)
- Applying Gauss' theorem to an open surface
- Applying Green's theorem without a suitable closed plane curve
- Wrong boundary orientation in Stokes' theorem
- Forgetting the outward normal in a flux integral
- Assuming zero curl ⇒ conservative without domain conditions